(Please write your Exam Roll No.)

Exam Roll No. .....

END TERM EXAMINATION

FIRST SEMESTER [B.TECH.] MARCH 2023

Paper Code: ES-119 **Subject: Manufacturing Process**

Time: 3 Hours **Maximum Marks: 75**

Note: Attempt five questions including Q.No.1 which is compulsory. Select one question from each unit.

Q1 a) What is Manufacturing? List out different manufacturing processes. [2]

b) What is the difference between pattern and casting? [2]

c) Explain the effect of polarity on penetration in DC arc welding. [2]

d) Explain the basic principle of resistance welding. [2]

e) What is the difference between open and closed die forging process? [2]

f) What is powder metallurgy? [2]

g) Define the term plastic. What are its characteristics? [2]

h) List out four types of patterns. [1]

UNIT-I

Q2 a) Describe in detail about the importance of manufacturing towards technological and social economic development. [7]

b) Describe properties of materials: Hardness, elasticity and malleability. [3]

c) What are the desirable properties of molding sand? [5]

Q3 a) Why is it important to provide a means of venting gases from the mold cavity? [3]

b) Describe the process of centrifugal casting with neat sketch. What are the advantages? [7]

c) List out the defects in casting process. Explain any three with neat sketch. [5]

UNIT-II

Q4 a) Describe the oxy-acetylene gas welding technique and give the applications. [7]

b) Explain the principle of submerged arc welding with the help of neat sketch. Discuss its applications. [8]

Q5 a) What are differences between TIG and MIG welding processes? [5]

b) List out the defects in welding process. Write causes and remedies of following welding defects: i) Spatter ii) Overlap iii) Undercut [7]

c) What is the difference between filler metal and flux materials? [3]

P.T.O.